OPERATOR AND ARCHITECTURE NOTE

Exposing Narration Studio to n8n

What was built on 22 August 2026, the Docker Compose additions for the existing n8n stack, and how the Windows job API, custom node, and form workflow fit together.

Companion docs/n8n-service.md · docs/NarrationStudio-Internal-Workings.pdf §19

Node package n8n/n8n-nodes-narration-studio

Workflow n8n/workflows/narration-studio-form.json

Date 22 August 2026

Secrets This PDF names keys and compose fields. It does not print passwords or API keys.

1. What transpired

Narration Studio was a Windows console factory: transcript → rewritten talk → N scripts → slides → MP3s → segment MP4s → merged `video_{RunId}.mp4`. Microsoft Agent Framework (MAF) is the inner agent runtime. This repo's **production harness** still owns I/O. All PNG / MP3 / MP4 writes go through `ProductionTools`.

The request was to expose that factory as a **service for n8n**, with a **custom node** whose fields come from `src/NarrationStudio/appsettings.json`, and file browse for the transcript and stub narration (not typed Windows paths).

n8n already runs in Docker on the same PC. The container cannot see `E:\...`. Studio must stay a Windows process so it can use ffmpeg, SAPI, and Ollama on `localhost:11434`. A full video run is longer than n8n's HTTP timeout, so the API is async (`202` + `jobId`). Jobs run one at a time under `output/jobs/{jobId}/`. API keys stay off the public form.

n8n Form Trigger — browser uploads + labeled defaults from appsettings.json

▼

Custom node — Start Job / Get Status / Wait / Download (n8n-nodes-narration-studio)

▼

Windows Kestrel :5088 — POST /v1/jobs → JobQueue → JobWorker

▼

ProductionHarness — same loop as the CLI, scoped options overlay

▼

ProductionTools — only I/O surface; artifacts under output/jobs/{id}/

Choice	Result
Where n8n runs	Existing Docker stack `n8n-api_icountera`. Studio is not containerized.
How Docker reaches Studio	`http://host.docker.internal:5088`
Browse file	Form upload → node embeds `transcriptText` / `stubNarrationText` → Studio writes job inputs
CLI vs service	Empty args (or `--service`) start Kestrel. `--orchestration` still runs the CLI and exits.
Community npm	**Do not use** "Install community nodes" for this name. That hits the public registry, not this repo.

2. Docker Compose file (this PC)

Live file: `E:\icountera\n8n-api_icountera\n8n\docker-compose.yml`. Sanitized copy in this repo (ADDED lines commented): `n8n/docker-compose.n8n-excerpt.yml`. Do not paste `.env` passwords into tickets. Always compose with project name `n8n-api_icountera`.

> **DO NOT CREATE A SECOND STACK**
>
> `docker compose up` without `-p n8n-api_icountera` created another stack that collided on port 5678. That extra stack was torn down. Use the project flag every time.

2.1 What was already there vs what was added

Already on the stack	Added for Narration Studio
`postgres:16`, volumes `n8n_data` / `postgres_data`	`N8N_CUSTOM_EXTENSIONS: /custom-nodes/n8n-nodes-narration-studio`
`docker.n8n.io/n8nio/n8n:latest` on port **5678**	`N8N_UNVERIFIED_PACKAGES_ENABLED: "true"`
Postgres + `N8N_ENCRYPTION_KEY` from `.env`	`extra_hosts: host.docker.internal:host-gateway`
Existing n8n env (host, webhook, runners, …)	Volume mount of this repo's node package (read-only)
	One-time `npm install` under `/home/node/.n8n/nodes`

2.2 n8n service after the addition

```
n8n:
  image: docker.n8n.io/n8nio/n8n:latest
  restart: unless-stopped
  ports:
    - "5678:5678"
  environment:
    # … existing DB / encryption / host keys unchanged …
    N8N_CUSTOM_EXTENSIONS: /custom-nodes/n8n-nodes-narration-studio   # ADDED
    N8N_UNVERIFIED_PACKAGES_ENABLED: "true"                          # ADDED
  extra_hosts:
    - "host.docker.internal:host-gateway"                            # ADDED
  volumes:
    - n8n_data:/home/node/.n8n
    - E:/exercises/maf/n8n/n8n-nodes-narration-studio:/custom-nodes/n8n-nodes-narration-studio:ro  # ADDED
  depends_on:
    postgres:
      condition: service_healthy
```

On another machine, change the left-hand volume path to that clone's `n8n/n8n-nodes-narration-studio` folder.

2.3 Apply, install, restart

```
cd E:\icountera\n8n-api_icountera\n8n
docker compose -p n8n-api_icountera up -d
docker compose -p n8n-api_icountera exec n8n sh -c "mkdir -p /home/node/.n8n/nodes && cd /home/node/.n8n/nodes
&& npm install /custom-nodes/n8n-nodes-narration-studio"
docker compose -p n8n-api_icountera restart n8n
```

2.4 Update the Narration Studio node

The volume is live (writable on Windows, read-only in the container). Edit `n8n/n8n-nodes-narration-studio/dist/` in this repo.

Change	What to run
JS / credentials / SVG only	`docker compose -p n8n-api_icountera restart n8n` then **Ctrl+Shift+R**
`package.json` name / files / `n8n` block	`npm install` again (same exec line), then restart + hard-refresh
Workflow JSON	Re-import `n8n/workflows/narration-studio-form.json`

2.5 Add another custom node to the same compose

Each n8n package is its own folder with a `package.json` `n8n` key (same shape as `n8n-nodes-narration-studio`). This is an **n8n node**, not a MAF specialist — new Narration Studio agents still go in `src/NarrationStudio/Agents/` and `ProductionTools`.

1. Create `E:\exercises\maf\n8n\n8n-nodes-<name>\`.
2. Add a volume and append the path. In the Linux container, separate `N8N_CUSTOM_EXTENSIONS` folders with a **colon**:

```
      N8N_CUSTOM_EXTENSIONS: /custom-nodes/n8n-nodes-narration-studio:/custom-nodes/n8n-nodes-<name>
    extra_hosts:
      - "host.docker.internal:host-gateway"
    volumes:
      - n8n_data:/home/node/.n8n
      - E:/exercises/maf/n8n/n8n-nodes-narration-studio:/custom-nodes/n8n-nodes-narration-studio:ro
      - E:/exercises/maf/n8n/n8n-nodes-<name>:/custom-nodes/n8n-nodes-<name>:ro
```

```
cd E:\icountera\n8n-api_icountera\n8n
docker compose -p n8n-api_icountera up -d
docker compose -p n8n-api_icountera exec n8n sh -c "cd /home/node/.n8n/nodes && npm install /custom-nodes/n8n-
nodes-<name>"
docker compose -p n8n-api_icountera restart n8n
```

Hard-refresh n8n. Do **not** mount the whole `n8n/` folder as custom extensions — it also contains `workflows/` and this excerpt, which are not node packages.

2.6 What the stack is

Item	Value
Compose project	`n8n-api_icountera`
Compose dir	`E:\icountera\n8n-api_icountera\n8n`
n8n UI	`http://localhost:5678`
Studio listen	`http://0.0.0.0:5088` on Windows
From n8n → Studio	`http://host.docker.internal:5088`
Install location	`/home/node/.n8n/nodes` (local npm install of the mounted folder)

Allow port **5088** in Windows Firewall if the container cannot connect. Do not point Ollama at a URL inside the n8n container. Studio talks to Ollama on the Windows host.

3. Run the Windows job API

From the **repo root** (empty args — do not pass `--orchestration`):

```
cd E:\exercises\maf
dotnet run --project src\NarrationStudio
```

Or F5 the `NarrationStudio` launch profile. Listen URL is `Service:ListenUrl` (`http://0.0.0.0:5088`). `dotnet run` binds the **bin** copy of `appsettings.json`.

> **ONE PROCESS ONLY**
>
> A second `dotnet run` fails with MSB3027 because `NarrationStudio.exe` is locked. Stop the listener on port 5088 before rebuilding.

CLI production is unchanged:

```
dotnet run --project src\NarrationStudio -- --orchestration direct
```

`Service:ApiKey` empty → unauthenticated API (warning in the log). Clients send `X-Api-Key` when set. How to mint and wire a key: §3.1.

3.1 Job API key (optional)

There is no generate-key endpoint. Mint a random string, store the **same** value on Studio and in the n8n credential, then restart the service. The custom node's credentials stay optional until you set a key. `/health` never requires it. Other routes then need header `X-Api-Key`. A mismatch is **401**.

1. Create a secret (PowerShell):

```
[Convert]::ToBase64String((1..32 | ForEach-Object { Get-Random -Maximum 256 }) -as [byte[]])
```

A GUID is enough for local use: `[guid]::NewGuid().ToString("N")`.

2. Put it on Studio. `dotnet run` binds the **bin** copy `src/NarrationStudio/bin/Debug/net10.0-windows/appsettings.json`:

```
"Service": {
  "ListenUrl": "http://0.0.0.0:5088",
  "ApiKey": "paste-the-secret-here"
}
```

Prefer not committing it. Keep source `ApiKey` empty and set an environment variable before `dotnet run`:

```
$env:Service__ApiKey = "paste-the-secret-here"
```

Stop and start Studio so the new value loads.

3. Put it on n8n. This is only n8n storing a secret and attaching it to the three Studio nodes. It is not a form field.

A **Narration Studio API** credential is one vault record:

Field on the credential	What it is	What Studio does with it
Base URL	Where Docker n8n calls Windows	Default `http://host.docker.internal:5088` — the host's port 5088, not `localhost` inside the container
API Key	Same string as `Service:ApiKey`	HTTP header `X-Api-Key` on job calls. Not required for `/health`
ElevenLabs / OpenAI / Azure API Key	Vendor keys	Other headers (`X-ElevenLabs-Api-Key`, …). **Not** `Service:ApiKey`

Create it: n8n → **Credentials** → **Add credential** → search **Narration Studio API** (from the mounted custom node, not public npm). Set Base URL to `http://host.docker.internal:5088`. Set **API Key** to the exact Studio string (no extra space or quotes). Leave vendor keys blank unless you want those headers. Save. **Test** hits `GET /health` and only proves the URL is reachable, not that the job key is correct.

Attach it on all three nodes that call Studio. The sample workflow is Form → **Start job** → **Wait until complete** → **Download merged video**. Open each of those last three, set **Credential** to the record you saved, and save the workflow. If only Start job has it, Wait/Download get **401**. **On form submission** does not get this credential.

On each request the node uses the credential Base URL (or falls back to `http://host.docker.internal:5088`) and, if **API Key** is non-empty, sends `X-Api-Key`. Studio compares that header to `Service:ApiKey` with an exact match. Empty Studio key → any request is allowed. Set Studio key + missing/wrong header → **401** `{ "error": "Missing or invalid X-Api-Key." }`.

Do not put `Service:ApiKey` on the Form Trigger. That URL is shareable; the job key is a server-to-server secret (n8n encrypts credentials with `N8N_ENCRYPTION_KEY`). Segment count, Ollama model, voice id stay on the form.

Putting an ElevenLabs key in **API Key** will not unlock TTS, and will 401 the job API if it does not match `Service:ApiKey`. After you change the Studio key, restart `dotnet run` and update the n8n credential. You do not

need to re-import the workflow unless a node still has no credential selected.

4. Job API

Method	Path	Notes
GET	`/health`	No API key
POST	`/v1/jobs`	JSON or multipart. Returns **202** `{ jobId, status: queued }`
GET	`/v1/jobs/{id}`	`queued` / `running` / `complete` / `failed`
GET	`/v1/jobs/{id}/events`	`events.jsonl` lines
GET	`/v1/jobs/{id}/artifacts/merged`	MP4 download

Optional create headers (never form fields): `X-ElevenLabs-Api-Key`, `X-OpenAI-Api-Key`, `X-Azure-OpenAI-Api-Key`.

JSON may send booleans and ints as strings (`"true"`, `"8"`). `undefined` / `null` strings are ignored. Bad JSON is **400**, not 500. Files may arrive as `transcriptText` / `stubNarrationText`, or as multipart names `transcript`, `Transcript_file`, `stubNarration`, `Stub_narration_file`, `Dub_narration_file`.

Host fallbacks: `voice_transcript.txt` and `voice_transcript_ollama.txt` at the repo root when no upload is sent.

```
curl http://localhost:5088/health
curl -X POST http://localhost:5088/v1/jobs -H "Content-Type: application/json" ^
  -d "
{\"orchestration\":\"direct\",\"useStubNarration\":true,\"transcriptText\":\"hello\",\"stubNarrationText\":\"hello
stub\"}"
```

5. Custom node and form

Package `n8n/n8n-nodes-narration-studio` is shipped as JS in `dist/` (no TypeScript build). Node versions 2 and 3 share the same code. Operations: **Start Job**, **Get Status**, **Wait Until Complete**, **Download Merged Video**.

Credentials (**Narration Studio API**) are optional. Default base URL is `http://host.docker.internal:5088`. Put ElevenLabs / OpenAI / Azure **keys** on the credential, not the form.

Workflow `n8n/workflows/narration-studio-form.json`: Form Trigger → Start job → Wait until complete → Download merged video. Every field has a label and a hint with the config key. Defaults match `appsettings.json`. File fields cannot be pre-filled.

On the form: title, orchestration, stub flags, transcript and stub uploads, Model endpoint/model/timeout, segment/slide/talk caps, OpenAI/Azure model names, ElevenLabs voice/model ids, video size, rewrite extra instructions, keep artifacts.

Not on the form: API keys; `OutputDirectory` (always `output/jobs/{id}`).

Form Trigger names binaries after the field **label** (`Transcript_file`, `Stub_narration_file`). The node accepts those aliases.

6. What went wrong

1. **Public npm install** — n8n's "Install community nodes" dialog does not see this repo. Use the compose mount + local `npm install` in §2.
2. **"Node does not have any credentials set"** — credentials were made optional.
3. **Start job 500** — n8n sent `"useStubNarration": "true"` (string). Studio now accepts string bools/ints and embeds file text. Parse failures return 400.
4. **MSB3027 file lock** — two `dotnet run` instances. Stop port 5088 first.

End of document. 22 August 2026. If the code and this PDF diverge, the code wins. Markdown twin: docs/n8n-service.md.